

1.) Diket: $\Delta t = 10^\circ\text{C}$
 $Q_v = 3$ kali lipat
 $v_1 = 81\text{V}$
 $T_1 = 20^\circ\text{C}$

Dit: $T_2 = ?$

Jawab:

Suhu naik $T_2 > T_1$

$$v_2 = 3 \frac{T_2 - T_1}{10} v_1$$

$$Q_1 = 3 \frac{T_1 - 20}{10}$$

$$3^4 = 3 \frac{T_2 - 20}{10}$$

$$4 = \frac{T_2 - 20}{10}$$

$$40 = T_2 - 20$$

$$40 + 20 = T_2$$

$$60^\circ = T_2$$

2.) Diket: $\Delta t = 20^\circ\text{C}$
 $Q_v = 3$ kali lipat
 $T_1 = 30^\circ\text{C}$
 $T_2 = 90^\circ\text{C}$
 $t_1 = 3$ menit

Ditanya: $t_2 = ?$

Jawab: $T_2 > T_1$

$$t_2 = \frac{1}{\frac{T_2 - T_1}{3 \cdot 20}} t_1 = t_2 = \frac{1}{\frac{90 - 30}{20}} \cdot 3 = \frac{1}{3^2} \cdot 3 = \frac{1}{3^2} = \frac{1}{9} \text{ menit}$$

3.) Diket = $\Delta t = 10^\circ\text{C}$

$\Delta v = 2$ kali lipat

$T_1 = 20^\circ\text{C}$

$T_2 = 40^\circ\text{C}$

$t_1 = 40$ detik

Ditanya = $t_2 = ?$

Jawab

$T_2 > T_1$

$$t_2 = \frac{1}{2^{\frac{T_2 - T_1}{10}}} \cdot t_1$$

$$= \frac{1}{2^{\frac{20}{10}}} \cdot 40$$

$$= \frac{1}{2^2} \cdot 40$$

$$t_2 = \frac{40}{2^2} = 10 \text{ detik}$$

✓

4.) Cara mempercepat laju reaksi:

1. memperluas bidang sentuh
2. menaikkan suhu atau temperatur
3. katalis (~~men~~ menurunkan energi aktivitas)
4. menambahkan konsentrasi pereaksi